

Wenxin Chen

cwx@umich.com · github.com/legallchaperone · Ann Arbor, MI (starting Fall 2026)

Education

University of Michigan, Ann Arbor
B.S.E., Computer Science and Engineering

Sep 2026 - Expected May 2028

Shanghai Jiao Tong University - University of Michigan Joint Institute
B.Eng., Electrical and Computer Engineering

Sep 2024 - Aug 2026
GPA: 3.81/4.0 | Top 10%

- **Relevant Coursework:** MATH186 Honors Mathematics (A+), MATH214 Linear Algebra, EECS280 Data Structures & Algorithms (A), MATH256 Applied Honors Calculus IV (A), ECE4010J Probabilistic Methods in Engineering (A+)

Publications

- W. Gu, T. Luo, Q. Zhang, M. Ye, X. Shen, **W. Chen**, Y. Li, Y. Zhang, J. Hong, B.-L. Lu, W.-L. Zheng. “**CerebraGloss: Instruction-Tuning a Large Vision-Language Model for Fine-Grained Clinical EEG Interpretation.**” *Accepted, ICLR 2026.*
Key result: A 3B-parameter instruction-tuned VLM outperforms GPT-4o by **+10 points** on MCQ accuracy for clinical EEG interpretation.
- W. Gu, **W. Chen**, B.-L. Lu, W.-L. Zheng. “**BrainTune: An Artifact-Aware Meta-Contrastive Framework for Fast and Lasting Personalization of EEG Emotion Models.**” *Accepted, CogSci 2026.*

Research Experience

BCMI Lab, Shanghai Jiao Tong University
Undergraduate Research Assistant | Advisor: Prof. Wei-Long Zheng

May 2025 - Dec 2025

Project 1 - CerebraGloss

(ICLR 2026)

- Designed a **multi-task instruction dataset pipeline** that converts raw EEG waveforms and structured clinical metadata into diverse instruction-following samples (description, open-ended QA, MCQ), enabling instruction tuning of a 3B VLM.
- Built a unified **training, inference, and evaluation framework** with reproducible protocols; conducted ablation studies to identify critical design choices.
- Constructed **CerebraGloss-Bench**, an open-ended benchmark measuring generalization and fine-grained clinical interpretation quality of VLMs.
- Demonstrated that targeted **instruction tuning on domain-specific data** enables a 3B model to surpass GPT-4o by 10 points on MCQ accuracy, highlighting the gap between scale and task-specific supervision.

Project 2 - BrainTune

(CogSci 2026)

- Developed an **artifact-aware meta-contrastive learning** framework for rapid personalization of EEG emotion models under strong individual variability.
- Designed a **deployment-compatible pipeline** reducing offline preprocessing dependence; validated across multiple datasets and cross-session settings.

Technical Skills

Languages	Python, PyTorch, C/C++, MATLAB, L ^A T _E X
Tools	Git, Linux, Bash, Docker, HuggingFace Transformers
ML/AI	LLM Instruction Tuning, Vision-Language Models, Transformer Architectures, Diffusion Models, Reinforcement Learning (foundations)
English	TOEFL 115/120 (Speaking 29/30)

Other Experience

- Global College Advising Center - SJTU-UM Joint Institute** 2025 - Present
- Serve as student academic advisor; organize workshops on college application and academic planning, and hold walk-in office hours providing individualized advising to prospective and current students.
- Teaching Assistant - ECE2150J Intro to Circuits** May 2026 - Aug 2026
- Supported labs and recitation classes; graded homework; held office hours.
- Teaching Assistant - ENGL2000J Academic Writing II** Mar 2026 - May 2026
- Held office hours and provided writing tutoring.
- Teaching Assistant - MATH186 Honors Mathematics I** Fall 2025
- Held office hours and provided learning support for 90+ students in an honors calculus/analysis course.
- Student Union, Technology Division - SJTU-UM Joint Institute** 2024 - 2025
- Organized technical workshops (Linux, VSCode) and received Outstanding Member award.

Honors & Awards

- University Physics Competition (UPC) - Team Gold Medal Fall 2024
- Outstanding Member, Student Union Technology Division 2024-2025